

1.3 Vector Equations

Vector: ordered list of numbers.

Vectors in $\mathbb{R}^2$

A matrix with only one column is called a **column vector**, or simply a

vector.

Examples of vectors with two entries are $u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ where w_1 and w_2 are any real numbers.

The set of all vectors with two entries is denoted by $\mathbb{R}^2$ (read "R-two"). The $\mathbb{R}$ stands for the real numbers that appear as entries in the vectors, and the exponent 2 indicates that each vector contains two entries.

Two vectors in $\mathbb{R}^2$ are **equal** if and only if their corresponding entries are equal. Thus $\begin{bmatrix} 4 \\ 7 \end{bmatrix}$ and $\begin{bmatrix} 7 \\ 4 \end{bmatrix}$ are not equal, because vectors in $\mathbb{R}^2$ are **ordered pairs of real numbers**.